1 Progress Summary

This initial phase established the foundational infrastructure and validation necessary for subsequent machine learning analysis. Two essential components were completed: establishing the Euclid-Gaia cross-matching methodology, and validating which Euclid morphological classifiers can serve as reliable reference standards for training.

Euclid-Gaia cross-matching: Cross-matching was established between Euclid (ERO and Q1) and Gaia DR3 catalogues. Q1 sources were matched using `Gaia_id` identifiers; ERO sources via position-based matching (`m.v_alpha_j2000`, `m.v_delta_j2000`). Optimal matching radius was determined through photometric consistency analysis and Rayleigh distribution fitting.

Euclid classifier validation: Using Euclid ERO data, the `class_star` and `spread_model` classifiers were validated across $10 < G < 27$ mag. `spread_model` exhibits systematic magnitude-dependent bias rendering it unreliable, while `class_star` provides robust classification across all magnitude ranges. This establishes `class_star` as the reference standard for subsequent Gaia source classification.

2 Preliminary Findings

2.1 Euclid ERO Classifiers: `class_star` vs `spread_model`

Sanity check was performed for both Euclid ERO classifiers across $12 < G < 28$ mag. The `class_star` classifier (left panels) shows the expected monotonic behavior: point-like sources dominate at bright magnitudes, with extended fraction increasing smoothly toward fainter magnitudes. In contrast, `spread_model` (right panels) exhibits a characteristic bumpy structure caused by magnitude-dependent systematic effects.

The root cause was that `spread_model` classification uses an error-weighted threshold that depends on its own the uncertainty `spreaderr_model`, where the threshold for star-galaxy separation is calculated by

$$\sqrt{\epsilon^2 + (\kappa \times \texttt{spreaderr_model})^2}$$

where ϵ and κ are constants. Since `spreaderr_model` is inversely related to SNR, this creates magnitude-dependent bias:

- **Very bright regime** ($G < 15$ mag): Noisy region where `spread_model` shows erratic behavior with high scatter.
- **Bright regime** ($15 < G < 19$ mag): High SNR measurements produce low `spreaderr_model`, thus tight astrometric threshold for star-galaxy separation. Genuinely point-like sources with small positive `spread_model` values exceed this tight threshold and are misclassified as extended, producing the first bump.
- **Intermediate regime** ($19 < G < 22$ mag): Temporarily reasonable classification where SNR-dependent threshold effects balance out, but this narrow range is insufficient for general use.
- **Faint regime** ($G > 22$ mag): Low SNR produces high `spreaderr_model` values, loosens the threshold. Extended sources with noise-inflated `spread_model` values fall below this loose threshold and are misclassified as point-like, producing the second bump.

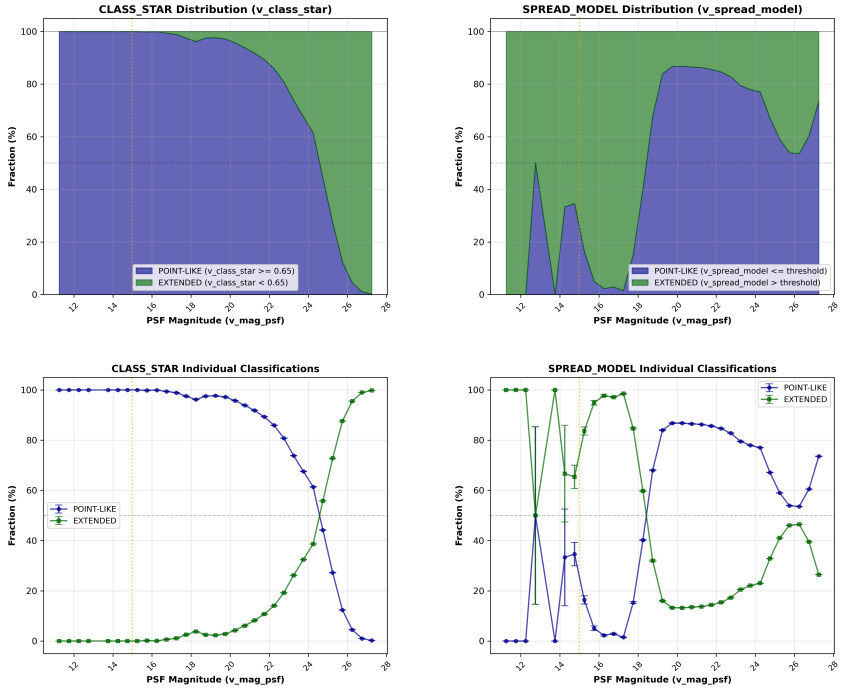


Figure 1: ERO classification distribution vs magnitude. Top: Stacked area plots showing point-like (blue) and extended (green) fractions. Bottom: Individual classification fractions with binomial uncertainties.

This error-weighted threshold approach, while intended to maintain consistent classification reliability across magnitude ranges, instead introduces systematic bias by making the classification criterion itself magnitude-dependent rather than morphology-dependent, rendering it unreliable across magnitude ranges. Thus we determined classifier `class_star` is preferred over `spread_model` for extended source identification.

2.2 Q1 Photometric Classification

PHZ classifications show expected trends (stars at intermediate magnitudes, galaxies at faint end) at faint end ($G > 22$ mag), but suffer from classification uncertainty at bright end ($G < 22$ mag)

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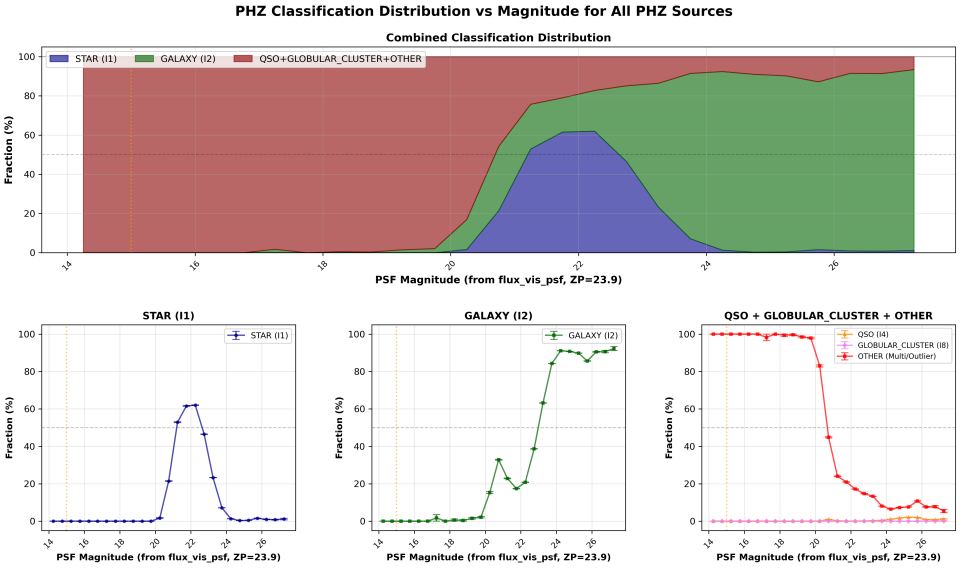


Figure 2: PHZ classification distribution vs magnitude. Top: Stacked area plot showing STAR (I1, blue), GALAXY (I2, green), and combined QSO+GLOBULAR_CLUSTER+OTHER (red) fractions. Bottom: Individual classification fractions.

3 Cross-Matching Methodology

3.1 Optimal Matching Radius Determination

Angular separation distributions between Q1 and Gaia DR3 matched sources were analyzed to determine optimal matching radius. The distribution in Figure 3 exhibits a tight core with median separation of 0.093 arcsec and long tail extending to ∼10 arcsec, characteristic of a genuine match population with spurious association contamination at large separations.

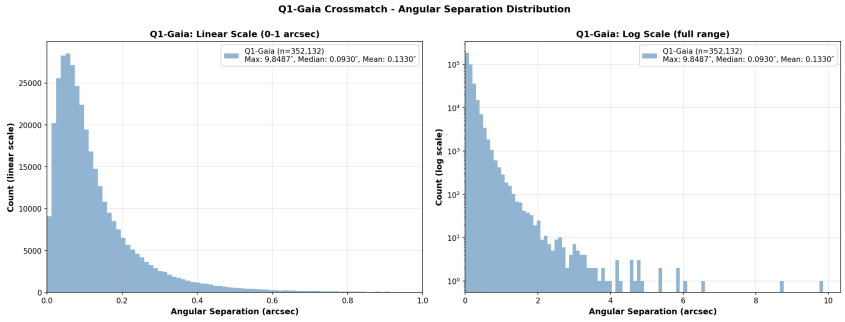


Figure 3: Q1-Gaia angular separation distribution. Left: Linear scale showing tight core with median 0.093 arcsec. Right: Log scale showing full distribution extending to ∼10 arcsec.

Rayleigh distribution fitting was performed to statistically determine the matching radius, as this model applies when positional uncertainties in RA and Dec are independently Gaussian. The fit yields scale parameter $\sigma = 0.177$ arcsec (Figure 4). Statistical tests show deviations from ideal Rayleigh behavior (KS statistic = 0.188, $p < 0.001$), likely reflecting known systematic astrometric effects in Gaia and Euclid catalogues, such as parallax zero-point offsets, proper motion systematics, and catalog-specific distortions, that could cause discrepancies from perfectly independent Gaussian errors.

Despite imperfect fit, the Rayleigh model adequately describes the core distribution and provides a practical statistical estimate. The fitted scale parameter corresponds to a characteristic matching radius of $\sigma = 0.177$ arcsec, with 95% of the distribution within 0.44 arcsec. A matching radius of 0.17 arcsec (approximately 1σ) was selected for photometric validation, balancing completeness against contamination.

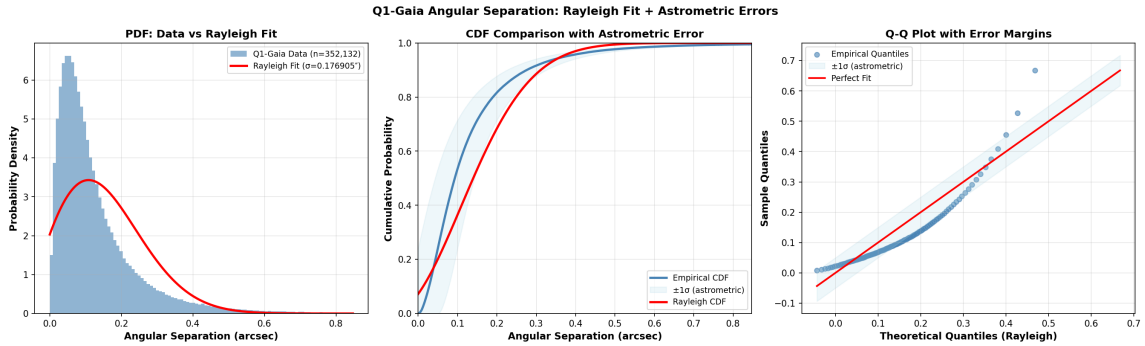


Figure 4: Rayleigh distribution fit to Q1-Gaia angular separations. Left: PDF comparison with fitted Rayleigh. Middle: CDF comparison showing systematic offset at large separations. Right: Q-Q plot showing deviation from theoretical quantiles.

3.2 Photometric Consistency Check

To validate the statistically-determined 0.17 arcsec matching radius, photometric consistency was examined at this radius along with tighter (0.05 arcsec) and looser (1.0 arcsec) comparisons. Figure 5 shows magnitude differences as a function of angular separation for both Q1-Gaia and ERO-Gaia cross-matches.

The 0.17 arcsec radius captures 76.7% of Q1 sources and 85.5% of ERO sources while maintaining reasonable photometric consistency: magnitude differences show no systematic offset and minimal scatter compared to tighter matching. The tight 0.05 arcsec radius achieves higher purity but sacrifices completeness. The loose 1.0 arcsec radius achieves near-complete matching but includes significant contamination from spurious associations at large separations.

The 0.17 arcsec radius provides optimal balance: high completeness, photometric reliability, and minimal contamination. This radius was adopted for all cross-matching operations.

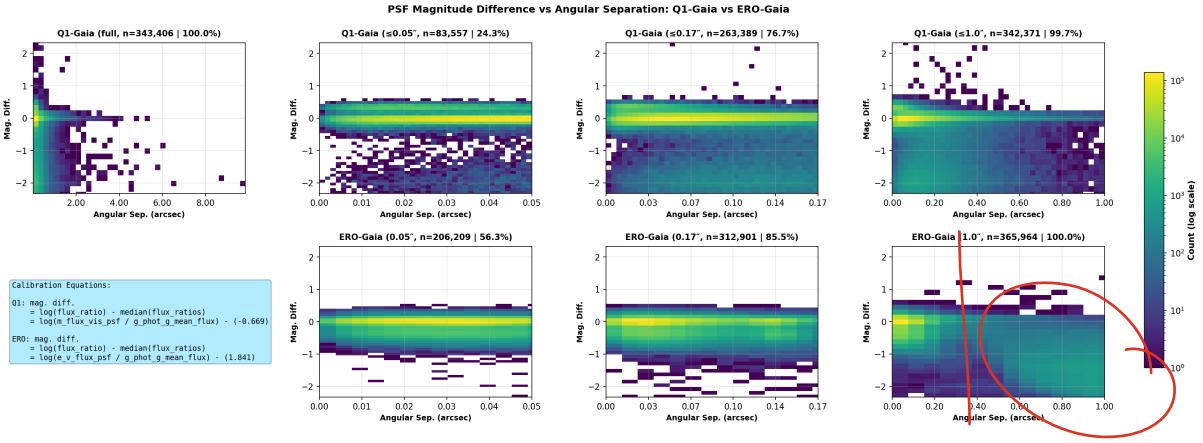


Figure 5: Photometric consistency check at multiple radii. Magnitude difference vs angular separation for Q1-Gaia (top row) and ERO-Gaia (bottom row) at full sample, 0.05, 0.17, and 1.0 arcsec. Color indicates log scale source density. Calibration equations shown in lower left.

4 Future Work & Timeline

4.1 Upcoming Analysis

Photometric Systematics: Characterize magnitude differences Δmag (Euclid − Gaia) as functions of magnitude, color (BP−RP), and morphological class to identify systematic photometric offsets. Investigate extended source distribution in color-magnitude space.

Machine Learning: Identify optimal Gaia parameters for machine learning, train XGBoost classifier using Euclid labels, and perform feature importance analysis to identify key diagnostic parameters.

Extended Source Characterization: Apply clustering methods (UMAP/self-organizing maps) for morphological sub-classification. Extract physical properties (size, surface brightness) from Euclid images and cross-validate classifications.

Full Catalogue Application: Deploy validated classifier to Gaia DR3, generate extended source catalogue with morphological types, compare population demographics with Euclid overlap regions, analyse spatial distribution, clustering properties, and physical interpretations by morphological type.

4.2 Project Goals

Primary objective: Develop and validate machine learning pipeline for identifying and classifying extended sources in Gaia DR3 using Euclid training data.

Deliverables: (1) Validated XGBoost classifier for Gaia extended source identification, (2) feature importance analysis identifying key Gaia morphological indicators, (3) morphological sub-classification scheme for extended sources, (4) Gaia extended catalogue with classifications.

If time allows: Detailed physical characterization using Euclid imaging for representative samples, investigation of environmental dependence of extended source morphologies, comparison with existing catalogues.